FINAL CONSOLIDATED RESEARCH & ENGINEERING SPECIFICATION

AICTE APPROVAL WORKFLOW

Integrated Approval Workflow Automation, Compliance Monitoring and Alert Management System

Smart India Hackathon 2026 • Problem Statement 26262 • AICTE Regulation Bureau

Research synthesis date: 22 October 2026 | Evidence scope: public official records + four supplied research reports

Status: implementation-ready prototype blueprint; NOT an AICTE-authorized system, production integration, or legal compliance certificate.

Decision in one minute

Build a policy-version-aware regulatory case platform that accurately routes different applications, restricts re-scrutiny to recorded deficiencies, coordinates EVC tasks, connects approval decisions to subsequent institutional obligations, and makes every timer or alert explainable through source-linked event history. Treat official government decisions as human-only actions. Use real published rules where verified and clearly labeled simulated policies for otherwise undocumented internal SLAs.

What is settled	What remains a dependency
AICTE has portal workflows; the 2024–27 APH and 2026–27 errata govern example rules.	Current officer-side screen coverage and internal escalation configuration.
Current intake-increase routes can be accreditation-conditional.	Ambiguous threshold conditions and any inaccessible public notices.
Re-scrutiny scope is specifically restricted for APH §1.8(g).	Exact UI-level delegation in AICTE authenticated services.
An end-to-end prototype can be implemented with fictional case events.	Real AICTE production APIs and authorized deployment.

Read first: Sections 1–4 establish the problem and official baseline. Sections 8–16 define exactly what to build. Sections 17–20 hand off the implementation and testing. Sections 23–24 preserve the source audit.

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PART I — Understanding the problem

1. Executive summary

PS 26262 is a regulatory coordination problem, not a request to invent online admissions or to replace AICTE officials. Institutions apply for approvals; the required processing route varies with the application type, academic session, accreditation eligibility and earlier committee outcomes. As cases move across scrutiny, re-scrutiny, EVC and possible hearing/appeal stages, managers need ownership, evidence, deadlines and reasons for every state change. Some approvals also generate obligations that remain active long after the application decision. [PS, A, B, C]

AICTE already publicly documents automated committee-member selection, deficiency communication, reports and official decision publication through its portal. It also publishes approval rules and later amendments. Therefore the proposal should improve coordination and explainability rather than claim that digital approval processing does not exist. [G01, G02, G03]

Recommended platform An administrative coordination layer organized around a versioned policy registry, guarded case state machine, differential re-scrutiny, officer-controlled decisions, obligation and lease lifecycle, deadline engine, cross-role dashboards and traceable audit events.

Three central differentiators (1) Source-linked, session-aware route explanations; (2) enforcement of the verified re-scrutiny scope restriction; (3) approval-to-obligation continuity with reasoned, exception-aware reminders. Small retrospective conformance checks reinforce these without making unvalidated AI claims.

What the prototype proves Two cases follow different routes; initial scrutiny flags D1/D2; the correction view changes only D1/D2; an invalid attempt on cleared C3 is rejected; an eligible EVC task is created; a human-approved demo decision creates a compliance obligation; simulated time advances and triggers one explainable alert; submission/reschedule cancels stale reminders; the dashboard and audit show the complete journey.

2. Problem statement explained for everyone

Imagine Institution North wants to open a new course while Institution South requests a simple change to its official name. Although both submit applications to AICTE, the checks can be different. North may need document scrutiny and an EVC, while a specified name-change case may need scrutiny but no EVC. If the platform treats them identically, it wastes committee effort or skips a required check. The correct route depends on the applicable handbook and the exact circumstances. [G01, A §4]

Now imagine an institution receives an approval subject to submitting further supporting evidence or maintaining relevant documents. The approval event ends one workflow but starts another: reminders, evidence review, possible renewal and eventual authorized closure. PS 26262 asks for a unified view of both phases. A missed document expiry or a pending EVC should become visible before follow-up depends on manual searching.

2.1 Six mandatory functional domains

Domain	What user needs	Demonstrable proof
A Workflow	Category-appropriate stages; guarded transitions; deficiency return; reasons	Different route per case; invalid step blocked
B Monitoring	Region/officer/stage pendency, ages, ownership and workload	Dashboard counters change from real events
C EVC	Eligible queue, schedule, completion and overdue visibility	One EVC created only when applicable
D Undertakings	Conditional obligation, evidence upload, review and closure	Approval creates an obligation
E Documents	Document versions, validity and renewal evidence	Renewal suppresses stale expiry notice
F Alerts/audit	Scheduled reminders, controlled escalation, reports, event log	One deduplicated alert with reason and actor

2.2 Actors, boundaries, outcomes

The institution uploads applications, corrections and compliance evidence. Scrutiny and EVC participants evaluate what their authorized mandate covers. Regional/Bureau users coordinate queues and monitor overdue work. A competent authority records the legally reserved decision; an ordinary application administrator cannot simply grant approval. Student teams must not map a software “admin” role to a statutory authority without evidence of delegation. [G01 §§1.8–1.10; G02 definitions]

In scope Approval workflow routing and administrative monitoring, stage/workload dashboards, EVC task coordination, obligations and documents, messages/escalations, audit and secure user roles.

Out of scope Automated legal approval/rejection, unauthorized access to AICTE infrastructure, independently authenticating lease title or bank settlement from uploaded files, issuing legally certified evidentiary records, and training predictive models on nonexistent official histories.

PART II — Government ecosystem and AS-IS process

3. Existing AICTE systems: documented versus unknown

The 2024–27 handbook describes web-portal-based automated selection for committees, online deficiency communication, availability of scrutiny/EVC reports, and publication of decisions. Historic manuals additionally show institution application states, an AICTE notification indicator, certain payment records and an NSWS hand-off. These are existing functions and cannot be described as new innovations. The existence of a “notification” icon does not prove that a complete role-based overdue escalation matrix exists; absence of an internal screenshot also does not prove that AICTE lacks one. [G01, G04–G07; B pp.3–5]

Capability	Evidence status	Positioning rule
Institutional online submission	Verified in official/historic materials	Baseline, not USP
Portal-driven member selection	Verified in APH procedural language	Reuse as documented context
Deficiency/report exchange	Verified in APH	Add explainability rather than reinvent upload
Notification icon	Historic official manual	Do not infer triggers/recipients
Online security deposit payment	Historic manual	Financial-verification integration not presumed
Officer-stage backlog, cross-role SLA engine	Not publicly established	Cannot claim existing or absent
Approval-to-lease unified operational queue	Not publicly established	Proposed integration, not first-ever claim
Public AICTE case-status write API	Not verified	Synthetic importer/mock adapter only

3.1 AS-IS approval process and its limitations

A simplified documented sequence for applicable new-institution cases is: application submitted → scrutiny → deficiencies and scope-bound re-scrutiny if necessary → EVC when prescribed → authorized review/decision → possible further

remedies. This is not a universal route; other application categories may follow a self-disclosure or scrutiny-only path. Annexure 7 and Tables 2.2/2.3 provide the official baseline. [G01 §§1.8–1.11, Annexure 7, Table 2.3]

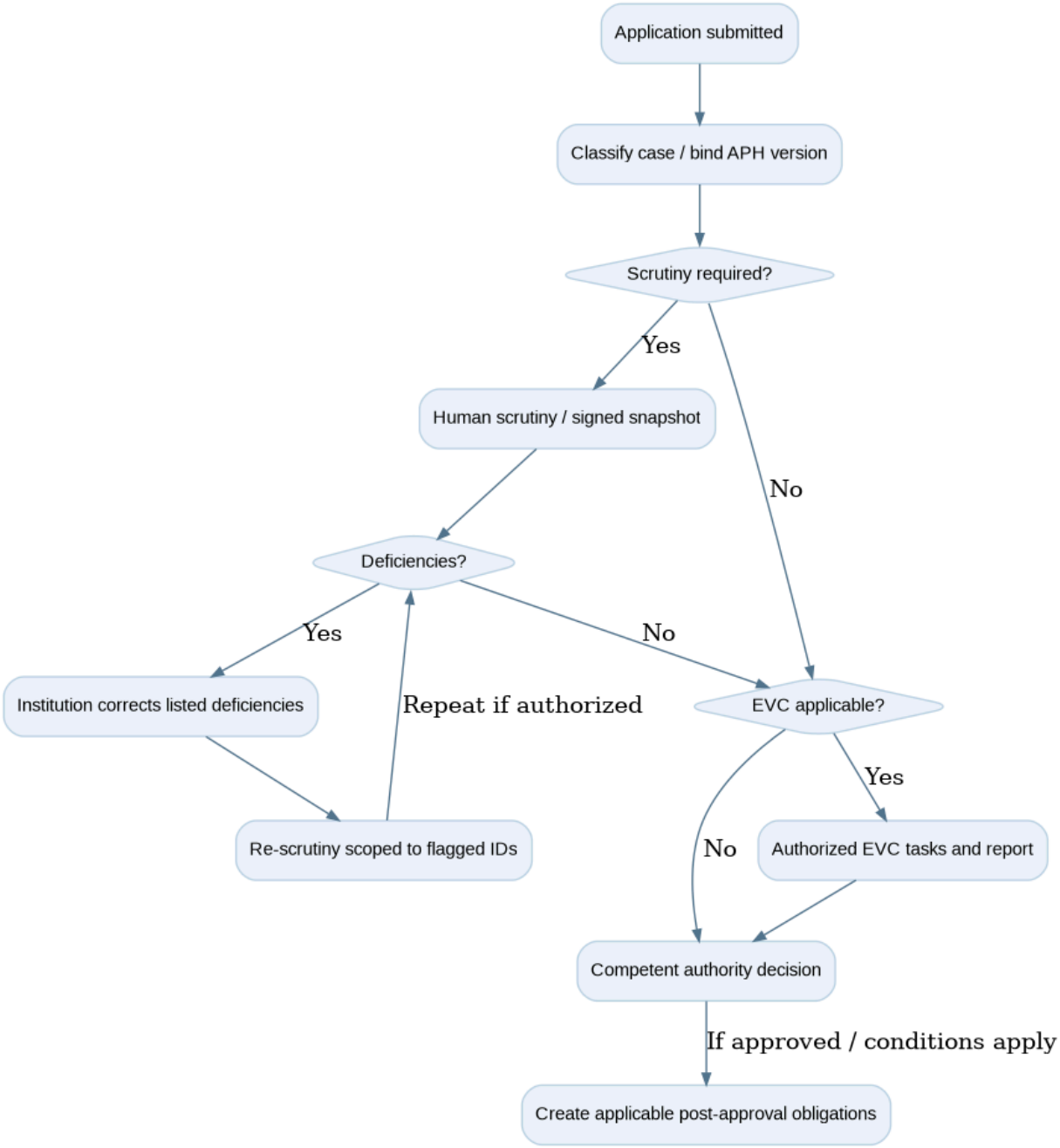


Figure 1. Proposed explanatory case flow; conditional branching depends on current application-specific rules. Human decisions remain mandatory.

3.2 What is an evidence-backed gap?

It is proven that official workflows differ and change, that re-scrutiny has a stated scope, and that some approvals create continuing obligations. It is also a problem-statement-identified need to integrate alerts, dashboards and document validity. The public source set does NOT prove that AICTE currently lacks all corresponding internal tools. The defensible gap is a proposed traceable integrated product addressing these requirements, with present-day competitive novelty conditional on an authorized system walkthrough. [A §6; B §§4–9]

4. Regulatory rules and current applicability

The binding starting point is APH 2024–2027 as modified by published addenda/corrigenda. The 42-page 2026–27 updated addendum contains a final-page errata (PDF page 42) and revised committee definitions. The rule registry must treat an original clause, later modification and final errata as separately identified source versions and must evaluate scope/effective period before choosing an active rule. Other unamended APH provisions remain potentially applicable. [G01–G03]

4.1 High-impact verified corrections

Regulatory issue	Current decision for software	Source
Committee constitution	Council constitution in amended definitions for EVC/SAC/SCSC/SHC; scheduling rights are a separate question.	G02 PDF pp.2–3
SHC applicability	"Wherever applicable"; never force SHC for every deficiency.	G02 PDF p.11 §2.23.3
Intake increase	Some evaluated routes differ for qualifying valid NBA accreditation versus non-accredited situations; do not make EVC universal.	G02 PDF p.42 §2.6(c)(i)
Exact X=360 intake value	Displayed table separates X < 360 and X > 360; do not silently infer the equality rule.	G02 PDF p.42
New January session	Published academic dates are session cutoffs, not arbitrary internal stage SLAs.	G03 Table D
Security deposit/Affidavit 3	Specified 15-day period measured from intimation, plus calendar cutoffs; legal consequences require official decision.	G02 §2.25(f), PDF p.11
Existing institution appeal	Where §2.24 applies, one appeal period is measured from publication/upload of the relevant EC decision.	G01 §2.24, printed p.29

4.2 The conditional intake example

Consider two fictional eligible intake-increase applications for AY 2026–27. One has a valid applicable NBA accreditation; the other does not. The final errata distinguishes evaluation with scrutiny/re-scrutiny for the specified accredited pathway and scrutiny/re-scrutiny/EVC for the specified non-accredited pathway. The UI must display which source version selected the route and whether extra eligibility conditions were satisfied. A case exactly at a displayed boundary of X = 360 should enter "Policy review required" until another applicable authoritative clause resolves it. Do not reinterpret "n divisions" as "n seats" without a verified rule. [G02 PDF p.42]

4.3 Re-scrutiny is narrower than a second full inspection

APH §1.8(g), printed page 7, says re-scrutiny verifies only the deficiencies pointed out during initial scrutiny. A safe engineering expression is to freeze a signed initial checklist snapshot, bind deficiency IDs to the correction task, allow amendments only to that set, and reject other changes unless a distinct authorized exceptional process is established. This operationalizes a written rule; it does not establish that the existing portal lacks similar controls. [G01 PDF p.23; D pp.2–3]

4.4 Regulatory rules are not operational SLAs

The official 2026–27 academic calendar distinguishes technical institutions, PGDM/PGCM, ODL and January-session provisions. Its 31 May 2026 AICTE approval cutoff for specified categories is not an instruction to label every case "late" after a seven-day officer processing timer. Keep external academic cutoffs, category-specific application windows and per-case event timers as separate policy types. [G03]

4.5 Decisions and consequences

A system can compute that an undertaking is approaching its source-defined due date, notify an institution, display missing evidence and prepare an officer review packet. It cannot issue an official penalty, grant approval, withdraw approval or certify a document's legal status without the prescribed authorization and verification. Document upload is an assertion until reviewed; payment-reference entry is not independently verified settlement. [G01, G02; C pp.2–3]

5. Evidence-conscious process map and edge cases

Case situation	Required guard or action	Demonstration status
Simple self-disclosure category	No EVC generated by default under the seeded applicable route	MVP example

Case situation	Required guard or action	Demonstration status
Scrutiny-only modification	Require scrutiny review; avoid extra unsupported stages	MVP example
Scrutiny + conditional EVC	Generate task after recorded authorized prior completion	MVP example
Repeated deficiencies	Version snapshots and permitted deficiency set per round	MVP example
Conditional SHC/appeal	Create only after category-specific trigger and rights check	Advanced test
Simultaneous closure and opening	Dependent case blocked until required closure outcome is recorded	Optional; G01 §1.8(i)
Officer reassignment	Atomic ownership change with reason and audit event	Advanced test
Deadline extension or correction	Invalidate old reminder and recompute from confirmed event	MVP test
Rule changes during active cases	Keep pinned version; migrate only with authorized policy	Advanced test

PART III — Applied research and the actual gap

6. Research that influences the implementation

Select papers because their methods change an engineering design decision. The evidence set supports flexible workflows, safe change of running processes, correct event logs, conformance explanations and time patterns. Research papers are NOT sources for AICTE portal capability, which must be established separately by government documentation. [A pp.4–5; D pp.2–4]

ID	Research work	Direct use
P01	Declarative workflows: Balancing between flexibility and support — van der Aalst, Pesic & Schonenberg (2009)	Constraint-based branches and exceptions; use for case guard design; not drop-in code.
P02	Correctness criteria for dynamic changes in workflow systems—A survey — Rinderle, Reichert & Dadam (2004)	Keep active cases pinned to approved policy versions; controlled migration.
P03	Event log imperfection patterns for process mining — Suriadi, Andrews, ter Hofstede & Wynn (2017)	Design event validation and duplicate/out-of-order detection.
P04	Process Mining Manifesto — van der Aalst et al. (2012)	Define adequate event schema; postpone empirical bottleneck ML pending real data.
P05	Time patterns for process-aware information systems — Lanz, Weber & Reichert (2014)	Different clock types, fixed dates, time lags and validity windows.
P06	Visualization of Compliance Violation in Business Process Models — Awad & Weske (2010)	Explain violated workflow rules and offending event sequences.
P07	Conformance checking: a state-of-the-art literature review — Dunzer, Stierle, Matzner & Baier (2019)	A limited deterministic historical-log checker is appropriate.
P08	YAWL: yet another workflow language — van der Aalst & ter Hofstede (2005)	Loop/choice/cancellation patterns as validation checklist; full language optional.
P09	Change patterns and change support features in process-aware information systems — Weber, Reichert & Rinderle-Ma (2008)	Support controlled rule and process change rather than silent in-place edits.

P06 is a 2010 revised workshop-proceedings work (the underlying workshop was held in 2009). P07 is the 2019 ACM published contribution; its 2020 arXiv posting is a later version. The original Gemini comparative report clarifies both bibliographic distinctions. [D §7; P06–P07]

6.1 Build versus reuse

Technology	Value	Decision
React / TypeScript	Rapid institutional and officer UI with typed requests	Recommended frontend
FastAPI + Python or NestJS + TypeScript	Clear REST contracts, role checks and validation	Choose ONE team-familiar backend
PostgreSQL	Transactions, relational constraints, JSON policy snapshots	Recommended authoritative store
Flowable	Human BPMN/CMMN orchestration	Consider production evaluation, not mandatory for MVP
Temporal	Durable long-running application steps	Optional later when operation complexity warrants it
Open Policy Agent	Reusable advanced policy evaluation	Optional; simple guard service sufficient initially
bpmn-js	Useful workflow viewer/editor subject to license rules	Optional UI; ensure watermark compliance
PM4Py	Historical process-mining analysis	Defer without real representative logs

7. What genuinely remains to be improved

A good problem diagnosis must separate a demonstrated rule constraint from a documented problem-statement request and an internal-implementation unknown. The system design is justified by integration and traceability needs even without the unsupported accusation that no existing tool handles these functions. [A §6, B §9]

Evidence-backed need	Current source strength	Product response
Different case routes and amended rules	Direct APH and addendum evidence	Policy-aware route registry + cited explanation
Re-scrutiny limited to original deficiencies	Direct APH §1.8(g)	Deficiency IDs + server guard + history
Condition-dependent SHC/EVC stages	Direct addendum with exceptions	Configurable conditional stage gates
Ongoing obligations and document validity	PS explicit; some annexure sources	Approval-linked obligation and version history
Case ageing and overdue coordination	PS explicit; internal implementation unverified	Cross-role operational queue + explainable alerts
Official API/data availability	No public access contract confirmed	Adapter interface + synthetic tests
Predictive bottleneck accuracy	No official historical logs	Not an MVP claim

PART IV — Complete proposed solution

8. The product and its main operating model

Working name PravahReg / AICTE CaseFlow (proposed project labels only; not official AICTE brands).

Value proposition Understand exactly which stage or obligation is pending, who owns it, which policy applies and why the next administrative action is permitted.

The platform consists of a shared case timeline, a current-session policy source register, human-controlled stage tasks, a compliance-obligation ledger and an event-driven operational monitor. Its primary organizing object is a regulatory case. A case carries the academic session, application category, linked institution, approved policy version, current stage, responsibility, evidence and sequential audit history. When an authorized event occurs, guards check whether the next administrative stage is permitted; the resulting event updates queues and, if relevant, schedules or cancels timers. [A §11; B §10]

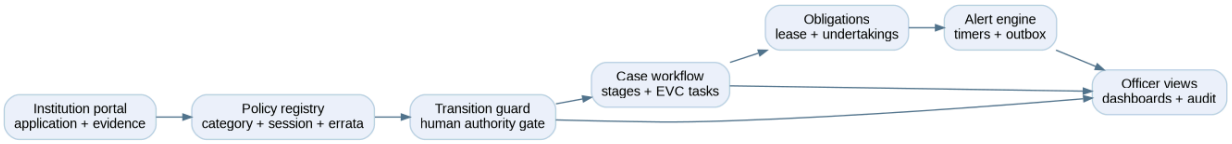


Figure 2. Proposed modular architecture. The policy registry is a decision-support component; the official authority acts through guarded human decisions.

8.1 Example complete case

1. A fictional institution submits an application for a category that requires scrutiny. The policy registry binds the applicable academic year and verified source clause.
2. An authorized scrutiny reviewer marks D1 and D2 deficient and signs a snapshot. The institution sees the two actionable observations and uploads corrected documents.
3. During re-scrutiny, the API permits D1 and D2 only. A forged update to cleared item C3 returns 403/409 according to the case guard and leaves an audit event.
4. After all required evidence is accepted, the route creates an EVC task only when the rule calls for it. Human EVC findings are uploaded and signed.
5. A competent authority enters the final decision. If the hypothetical approval carries a verified/simulated demonstration obligation, an obligation record is created with its source and exact due-date origin.
6. The deadline worker generates a single nonbinding reminder; an officer can see the trigger and reason. Replacement evidence changes the status; stale scheduled reminders are invalidated.
7. Region/Bureau dashboards aggregate pending case ages, upcoming tasks and compliance status; a timeline reconstructs every state-changing event.

8.2 Roles and workflow surfaces

User	Core actions	Prohibited/default restriction
Institution	Submit/update application, correct flagged deficiencies, submit compliance evidence, view its own case.	Cannot finalize scrutiny, alter rule versions or grant approval.
Scrutiny reviewer	Review assigned cases, issue signed observations, verify appropriate corrections.	Cannot change unflagged items in an ordinary re-scrutiny task.
EVC coordinator/member	View eligible queue, arrange authorized session/task, record evidence/report.	Cannot constitute committees or make final decision without delegation.
Regional/Bureau officer	Monitor queues, assign permitted tasks, log clarifications/authorised escalations.	Cannot exercise statutory decision authority based solely on an app role.
Competent authority user	Record verified official decision within applicable delegation and signed evidence.	No automated model substitutes for this role.
System policy administrator	Draft and test rule changes with required approval workflow.	Cannot silently activate a regulatory change or rewrite case history.

9. Basic system — full foundational capabilities

The basic system means everything required for a coherent and defensible product, whether or not a feature is included in the first prototype. It is distinct from the MVP and distinct from the USPs.

ID	Basic capability	Functional acceptance meaning
B01	Institution profiles and identifiers	Persist region, institutional identifiers, access mapping and status.
B02	Case intake	Capture application type, session, evidence and unique case reference.
B03	Policy registry	Version sources, applicability dates, reviewed rule conditions and supersession.
B04	Route selection	Select stage path by case facts and policy; expose reasons.
B05	Guarded stage transitions	Check authorized actor, required evidence, expected version and allowed next state.
B06	Deficiency and re-scrutiny	Preserve signed snapshot, deficiency IDs and corrected evidence.
B07	EVC/committee work items	Create eligible tasks, schedule state, completion and report reference.
B08	Authorized decisions	Separate official decisions from administrative state events.
B09	Undertakings and obligations	Record conditional duties, dates, evidence and review.

ID	Basic capability	Functional acceptance meaning
B10	Document validity	Store versions, status, expiry, renewal and verification details.
B11	Alerts and escalation	Create configurable event-driven reminders and controlled officer escalation.
B12	Role dashboards	Show institution/region/officer stages, ages, overdue and compliance status.
B13	Audit and reports	Record complete actor/time/source changes; export sanitized aggregates.
B14	Security and operations	Authentication, RBAC, secure files, backups, controlled configuration.

10. Complete feature catalogue

Features are individual product capabilities; describing one as a “feature” does not make it a USP. All features should be traceable to a problem or a reliability requirement.

Area	Core features	Operational enhancements / later scope
Institution experience	Guided intake, case status, deficiency corrections, evidence upload	Bulk templates, applicant action digest, accessible notices
Officer workbench	Assigned queue, checklist, return/revert, approval evidence	Saved filters, workload heatmap, delegated-review tracker
EVC	Eligibility queue, task state, schedule and report	Committee capacity modeling, calendar invitations subject to policy
Compliance	Obligation register, due states, institutional evidence, review	Recurring renewal packs, reviewer hand-off, historical obligations
Document registry	Document versions, validity date, reviewer decision	Dual-time knowledge history, comparison of replacement versions
Workflow integrity	Rule version, transition guard, deficiency IDs, event log	Conformance scan, case dependency graphs, rule-change preview
Alerting	Timer creation, pre-due/overdue notices, cancellation, inbox	Recipient matrix, channels, delivery metrics, policy audit
Analytics	Pendency by stage/region, age, task-owner counts	Cohort trends and forecasting only with representative logs
Admin/security	Role permissions, policy approval workflow, audit and exports	SAML/SSO and formal security review after authorization

10.1 Feature priority vs dependency

No institution compliance reminder should exist without a recorded obligation, applicable source and due-date policy. No EVC item should exist without a qualifying route and prior-step evidence. No re-scrutiny can execute before a signed initial deficiency snapshot. These dependency constraints are more important than the number of dashboard widgets.

11. Final MVP, distinct from the full system

The MVP is the smallest coherent journey demonstrating a real engineering improvement under valid assumptions. The initial functional prototype is a narrower time-boxed proof; no production AICTE integration is claimed.

MVP component	It must really work	Acceptance proof
Institution intake	Persistent case record from typed form or seeded CSV	After refresh, case and session remain
Versioned routes	At least two source-linked category routes	Same UI selects different applicable branches
Scrutiny/return	Officer can sign deficiency snapshot; applicant corrects	D1/D2 show history of updates
Scoped re-scrutiny	API rejects out-of-scope item mutation	Attempt to edit C3 fails and logs reason
EVC task	Generated only when route requires; report attached	Scrutiny-only case never creates EVC
Human decision event	Authorized demo actor records decision with evidence	Admin without role cannot approve

MVP component	It must really work	Acceptance proof
Obligation and document	Case generates a due obligation, upload/verify state	Separate submitted vs verified statuses
Timer and escalation	One clock-based due event; dedupe; cancellation	Advance simulated time; no stale duplicate
Dashboards and audit	Counters and event history reflect changes	Action, actor, date, rule source visible

Prototype exclusions Live AICTE login/API access; real committee constitution; real bank verification; official penalty/withdrawal; full process mining; temporal constraint solvers; legal evidence certificates; untrained predictive AI.

11.1 Screens and display actions

Screen	Who	Minimum content and interaction
Institution home	Institution	Submitted cases, action-required banner, upload/submit controls
New application	Institution	Category, academic session, institution reference, document upload
Application timeline	Institution/authorized officer	Stage cards, event history, next allowed action and reason
Scrutiny checklist	Scrutiny officer	Requirements, mark deficient, sign snapshot, return with reasons
Re-scrutiny workbench	Reviewer	Flagged items editable only; retained original checklist read-only
EVC queue	Officer/EVC	Case eligibility, owner, schedule, report status
Compliance board	Institution/official	Obligation due date, evidence state, renew/verify actions
Regulator overview	RB/Region	Pending/stage ageing, alerts, active cases, exceptions
Rule explanation	Authorized viewers	Active clause, revision, evaluated conditions and decision trace
Admin rules	Authorized policy owner	Draft/validate/approve policy version; no silent change

12. Final innovations and USP selection

Evaluate each USP against the public official baseline and existing enterprise software. The novelty is in a precise AICTE-oriented integration or enforcement of an existing obligation, not a claim of inventing workflow technology.

Candidate	Classification	Value and credible proof
Source-linked rule provenance + version pinning	PRIMARY USP	Show route reason, exact APH/addendum/errata, effective session and alternative path when facts differ.
Differential re-scrutiny scope guard	PRIMARY supporting USP	Recorded D1/D2 can change; C3 cannot be changed through ordinary re-scrutiny; invalid attempt explained.
Approval-to-obligation continuity	SUPPORTING USP	After human approval, new obligation, review task and relevant document appear on case timeline.
Exception-aware reminders	SUPPORTING engineering USP	Changed due date suppresses stale alert; explanation shows origin and state.
Small explainable event conformance scanner	OPTIONAL high-visibility enhancement	Detect skipped stage or unsigned completion in seeded history; human review only.
Time-pattern taxonomy	FOUNDATIONAL engine feature	Correct fixed/event-relative/validity time modeling; not a standalone novel engine.
Dual-time document provenance	LATER feature	Differentiate effective lease date from date reviewer first learned of it.
Predictive officer bottleneck AI	DEFER	No representative authorized logs or benchmark. Not safe to claim accuracy.
Basic portal upload, notifications, hybrid EVC	BASELINE, not USP	Already documented in public AICTE sources and common software.

Candidate	Classification	Value and credible proof
Court-ready hash-chain certificate	REJECT claim	Auditing can be tamper-evident in design; automatic legal admissibility is not established.

12.1 USP example 1: "Why this route?"

An officer opens a case requesting intake increase in AY 2026–27. The rule explanation displays: application category, valid accreditation evidence (or lack of it), applicable errata entry, route selected, and confidence/status of any unresolved boundary interpretation. A second similar case with changed accreditation conditions may legitimately require an EVC. The comparison makes regulatory reasoning visible rather than leaving a mysterious status. [G02 p.42]

12.2 USP example 2: re-scrutiny scope guard

Initial checklist shows D1 missing land document, D2 missing staff evidence and C3 already accepted. The applicant supplies corrected evidence for D1/D2. The authorized re-scrutiny task cannot overwrite the C3 conclusion; the server independently enforces the allowed deficiency IDs. The action log records whether a blocked update was attempted, without claiming that the tool independently prevents all legal disputes. [G01 §1.8(g); D §2]

12.3 USP example 3: one lifecycle of obligations

An officer enters an authorized approval decision with an attached condition requiring follow-up evidence. A linked obligation is created with a named responsible institution, specified proof, due-date origin, reviewer, and status. If it is fulfilled, its upcoming reminder is superseded; the application and obligation remain connected on the same timeline. The requirement is explicit in PS 26262, but the novelty compared with AICTE internal screens has not been verified. [PS; B §7]

PART V — Technical solution and consistency

13. System architecture and module boundaries

Recommended pattern Modular monolith + PostgreSQL + transactional outbox + separate scheduled worker. This minimizes coordination overhead for a prototype while allowing extraction into services if real workload justifies it.

Module	Inputs	Outputs and invariant
Identity and RBAC	Authenticated actor, institution/region scope	User can only see and alter authorized records
Policy registry	Rule JSON, source citation, effective session, approval status	Exactly one selected active applicable revision or review-required result
Case workflow	Current state, expected_version, proposed event, evidence	Atomic accepted transition or explicit rejection; no partial audit
Re-scrutiny guard	Signed checklist + deficiency ID set	Only flagged IDs accepted for ordinary rescrutiny
EVC tasks	Eligibility event, owner, committee record	Task created only after authorized prerequisite
Obligation ledger	Authorized outcome, applicable conditional rule	New obligation with verifiable source/trigger
Document registry	Uploaded file, document type, reviewer status	New version; previous record remains queryable
Deadline scheduler	Timer definition, start event, policy version	At most one active timer for dedupe key and state
Notifications/outbox	Committed domain event + recipient policy	Queued delivery event; retry idempotently
Audit/conformance	Committed events + active policy	Human-readable sequence, optional violation finding
Dashboards/reporting	Authorized aggregate queries	Counts reflect real stored event/case data

13.1 Why not a full BPM suite immediately?

Flowable and Temporal support long-running workflows and are valuable production evaluation candidates. However, a small two-route prototype can be more reliably completed as an explicit typed state machine with audited guards and durable PostgreSQL transactions. Choosing the smaller design is an implementation trade-off, not a claim that complex

workflow engines are unsuitable for AICTE. Reassess based on number of regulated process variants, long-running tasks, integration needs and operational staff.

13.2 Proposed event transaction

On transition request, server reads the application and current version, checks actor authorization and source-pinned route, validates evidence, inserts the transition and audit event in one transaction, increments the case version, and writes an outbox row. A separate worker publishes due alerts or dashboard updates. If two officers operate concurrently, one stale update must fail without emitting a second decision. [A §11; C §I4]

14. Data model and API contract

Entity	Essential fields	Important constraint
Institution	id, permanent_id, name, region, status	Scoped access and unique identifier
Application	id, institution_id, category, session, current_state, row_version, policy_version_id	Pinned policy; optimistic concurrency
PolicyVersion	id, scope, effective_from/to, source_id, approval_state, supersedes_id	Never silently overwrite active policy
StageTask	id, application_id, stage_type, assignee_id, entered_at, completed_at, status	Unique relevant active task as required
ScrutinySnapshot	id, case_id, checklist_version, signed_by, signed_at	Versioned immutable record in application UI
Deficiency	id, snapshot_id, field_id, reason, correction_status	Re-scrutiny API can mutate only authorized IDs
DecisionEvidence	id, case_id, decision_type, actor, evidence_ref, authorized_at	Decision separate from a generic state edit
EVCEvent	id, case_id, schedule_time, coordinator, report_ref, outcome	No EVC without applicable route
Obligation	id, case_id, rule_version, due_at, trigger_event_id, reviewer, status	Timer must have recorded origin
DocumentVersion	id, obligation_id, type, file_ref, valid_from, valid_to, received_at, verified_at, status	Document assertion distinct from independent verification
TimerJob	id, obligation_id, policy_ref, due_at, dedupe_key, state	Supersede/cancel on revision
AuditEvent	id, case_id, actor, action, before_state, after_state, occurred_at, source_ref	Append-only app event history; not automatic legal certification
Outbox	id, event_id, event_type, delivery_status, attempts	Transactional event publishing and idempotency
CaseDependency	parent_id, child_id, required_outcome, source_ref	Optional linked closure/opening prerequisite

14.1 Representative REST API surface

Method / path	Role and behavior	Failure behavior
POST /api/v1/applications	Institution creates application with category/session	422 invalid facts; 403 wrong institution
GET /api/v1/applications/{id}/explanation	Authorized viewer sees rule and route reasons	404 unauthorized or absent record
POST /api/v1/applications/{id}/scrutiny	Reviewer records signed checklist and deficiencies	409 outdated case_version
POST /api/v1/applications/{id}/rescrutiny	Reviewer submits corrections for authorized deficiency IDs	403 out-of-scope item; 409 stale version
POST /api/v1/applications/{id}/transition	Role-limited guarded event with expected_version	409 unmet prerequisite; 403 unauthorized
GET /api/v1/evc/queue	EVC/official retrieves eligible tasks	403 insufficient permission
POST /api/v1/evc/{taskId}/report	Assigned authorized member records report	422 absent evidence; 409 stage mismatch
POST /api/v1/applications/{id}/decisions	Competent authority records signed decision metadata	403 missing decision privilege
POST /api/v1/obligations/{id}/evidence	Institution uploads typed proof as unverified	413 oversized file; 415 unsupported MIME
POST /api/v1/obligations/{id}/verify	Authorized reviewer sets verification result	403 improper role; 409 superseded item

Method / path	Role and behavior	Failure behavior
GET /api/v1/dashboards/pendency	Region/Bureau filtered stage ages and due items	Only authorized scopes
GET /api/v1/applications/{id}/history	Chronological event log and reasons	Sensitive fields redacted per role

14.2 Example transition contract (illustrative, not official AICTE API)

POST /api/v1/applications/CASE-001/transition { "action": "complete_scrutiny", "expected_version": 8, "evidence_ids": ["EV-31"], "reason": "Signed checklist accepted" }

Response 200: { "case_id": "CASE-001", "new_state": "evc_pending", "version": 9, "event_id": "EVT-091", "policy_ref": "APH-2026-27-SCOPE-12" }. On invalid route return 409 with { "error": "PRECONDITION_FAILED", "failed_rule": "RULE-X", "message": "Signed scrutiny report required" }. These are internal example contracts, not AICTE services.

14.3 Document and data verification states

Use explicit states SUBMITTED, SOURCE_STATUS_OBSERVED, INDEPENDENTLY_VERIFIED, REJECTED and SUPERSEDED. An institution-uploaded file must never jump directly to independently verified on successful upload. Capture effective validity dates and receipt/verification dates separately; full bi-temporal querying may be phased later. Record file hashes for integrity checking if useful, but do not claim a legal certificate solely from a hash. [C §I3; D §5]

15. Correct workflow execution, timers, and recovery

15.1 State machine invariants

- Every application binds to one reviewed rule version at submission or on an authorized migration event.
- A transition is evaluated on server against permitted previous state, actor role, evidence and decision authority.
- A signed scrutiny snapshot anchors the allowed deficiency IDs for the corresponding re-scrutiny task.
- An EVC report can exist only within an eligible, authorized EVC task and cannot grant final approval by itself.
- An approval decision and any generated obligation are distinct, linked events with separate timestamps.
- An outbox notification may be delivered only if the underlying obligation/timer remains active.
- No duplicate stage completion or retry can generate two competing authoritative events.
- Appeal timing is category-specific and begins with the exact specified publication event where applicable.

15.2 Timer types and ambiguity handling

Policy type	Example	How modeled
Fixed date	Published academic application cutoff	Absolute calendar date; not per-stage SLA
Trigger-relative duration	15 days from valid official intimation	Trigger_event_id plus interval and source
Lag between activities	Completion window if authoritative policy specifies it	Two event IDs and allowed interval
Validity interval	Lease document from/to date	valid_from, valid_to plus verification status
Recurring obligation	Periodic evidence only when source applies	Schedule frequency tied to scope
Conditional activation	Alert starts only after authorized conditional approval	Guard expression + source
Extension/suspension	Authorized due-date extension	Original due date retained; new version + reason
Cancellation/supersession	New verified evidence received	Cancel obsolete unsent jobs and record reason

15.3 Failure and recovery

Use transaction boundaries for case state + audit event + outbox. Deliver notifications at least once with deduplication; on every send re-check obligation version/status. If delivery fails, retry with capped exponential backoff and a failed-delivery queue. Preserve UTC event timestamps and display local IST; compute calendar-based rules with the correct policy calendar and no invented working-day assumptions. On outages, reprocess the persisted outbox, not reconstructed UI events.

15.4 Historical conformance monitor (optional)

Run a small deterministic inspection of seeded legacy event histories. Detect (i) EVC completion when EVC was inapplicable, (ii) a stage completion missing a signed prerequisite, (iii) an ordinary re-scrutiny event changing an unflagged

checklist item, (iv) duplicate closure events, and (v) evidence submitted after closure without authorized reopening. Every alert links offending event IDs and rule revision, and queues review rather than changing official case status. [D §3; P06–P07]

16. Security, privacy, accessibility, and scalability

Use an authentication provider or session-based login with secure cookies, CSRF protections as applicable, short-lived sessions and strong password handling. Apply server-side RBAC, per-institution/region row filtering, deny-by-default access and explicit separation of policy author and policy approver. File uploads require size/MIME screening, storage paths outside web root, malware scanning where available, role-scoped download URLs, encryption in transit and sensitive-field redaction.

Security events require timestamps, actor/session IDs and reasons; the ordinary application role should not edit audit history. “Append-only” in a prototype means users cannot rewrite events through the UI/API; durable tamper evidence, archive retention, disaster recovery and courtroom admissibility are separate production requirements. No legal compliance guarantee is implied. [A §12; D §6]

Production concern	Prototype control	Scale-up action
Case consistency	Relational FK + optimistic row version	High availability DB, backups, transactions
Timer reliability	Persisted outbox + status checks	Durable queue, job observability and DLQ
Read dashboard latency	Indexed queries and materialized aggregates where needed	Capacity testing and partition strategies after evidence
Document security	Private object storage, role-scoped access	Retention/legal-hold policy after AICTE confirmation
Policy update safety	Draft → verified → approved → effective	Change approvals, formal version migrations
Multi-region	Institution and region scope in schema	Sharding/replication only with measured load
Accessibility	Keyboard flow, semantic errors, readable color contrast	WCAG audit and assisted-technology evaluation

PART VI — Build plan, QA, and SIH demonstration

17. Recommended stack and technical feasibility

Practical selection React + TypeScript + Vite; FastAPI/Pydantic (or NestJS if team has deeper TypeScript proficiency); PostgreSQL; SQLAlchemy or Prisma aligned to backend; private local/object storage; APScheduler/Celery where appropriate; a simple outbox worker; tests using pytest/Jest and Playwright.

Do not build both FastAPI and NestJS. Choose exactly one stack after a 30-minute team capability check. A modular state machine is sufficient for the prototype. Versioned JSON rules are usable only after creating narrow review and activation safeguards; a YAML file copied into an app is not itself regulatory compliance. Integrate with real AICTE APIs only after authorized technical documentation is received. [A §8–§11; B §8]

17.1 Estimated complexity and gating

Component	Relative effort	Readiness / risk
Simple institution intake + case list	Low	Can build using fictional cases
Version-pinned category routing	Medium	Requires carefully seeded, sourced routes
Signed checklist and D1/D2 re-scrutiny	Medium	Best implementation differentiator
EVC and obligation tasks	Medium	Use simplified schedules; no official committee API
Reliable timer/outbox	Medium	Use simulated clock and idempotent state transitions
Dashboards/audit	Low–medium	Query real records, no static mock counters
Conformance scanner	Medium	Optional if core journey already stable
Formal BPMN engine with dynamic migrations	High	Pilot/production evaluation
Predicted delay ML	High + data dependency	Defer without historical labels
Live AICTE production connectors	Authorization dependency	Use mock adapters only

18. Parallel execution strategy — six members

The ideal production specification is larger than a 48–72-hour prototype. The following delivery strategy is a constrained demonstration track, not a limit on research depth or on the future product. The critical path is policy fixture → route transitions → scoped re-scrutiny → EVC/obligation events → dashboard/demo tests.

Workstream	Owner	First deliverable	Integration dependency
W1 Product and UI shell	Member 1	Institution and reviewer screens, scenario navigation	Stable typed API contracts
W2 Case/workflow backend	Member 2	Application states, guarded transitions, rule pins	Policy JSON and DB migrations
W3 Scrutiny and EVC	Member 3	Deficiency snapshots, scope guard, EVC task/report	Case transitions/events
W4 Compliance/document engine	Member 4	Obligations, versions, verification states	Decision-event interface
W5 Timers/dashboard/integration	Member 5	Outbox alerts, dashboard aggregation, demo time control	Case and obligation events
W6 Research QA/pitch/integration	Member 6	Official source-to-rule cards, tests, deployment and video	All workstreams; no unverified claims

18.1 Staged delivery

Window	What to integrate	Minimum proof
Foundation / first development block	Repo, DB schema, seed 20 fictional institutions, 2–3 route fixtures, auth scaffolding	Case can be submitted and inspected
Second block	Transitions, scrutiny snapshot, deficiency-return loop, re-scrutiny API guard	Allowed D1 succeeds; forbidden C3 fails
Third block	Conditional EVC task + human decision + obligation creation	Different categories produce correct paths
Fourth block	Deadline worker, dedupe/cancel behavior, role dashboards, full audit	One realistic alert fires and resolves
Final integration	Automated tests, accessibility sweep, deployed demonstration, presentation/video	End-to-end demo reliably repeatable

Reserve submission time for running the official PPT template, PDF export, five-minute video capture, link permissions and final form verification. Prioritize a correct integrated journey over extra dashboard cards. If technical setbacks occur, freeze extra conformance and document dual-time features first; never remove the core route explanation, re-scrutiny guard or audit trace.

18.2 Deployment and data strategy

Use seeded fictional institutions, fictional staff accounts and clearly tagged simulated policies. Keep demonstration routes tied to specific official clauses only when interpretation is unambiguous. For internal, undocumented SLAs show a visibly fictional time policy—e.g., “Demo reminder after 48 simulated hours”—not a claimed AICTE statutory deadline. Reset the database to a known initial state before recording. Keep a local replay video/screen recording as fallback if the demo site fails.

19. Testing, acceptance catalogue, and measurable results

A strong report lists tests to execute; it does not invent pass counts or measured AICTE performance. Each scenario below has a deterministic expected result suitable for automated checks. “Source-linked” means the case shows the precise applicable rule source when one exists.

ID	Scenario / initial condition	Stimulus	Pass criterion	Priority
T01	Scrutiny-only category	Application submitted with valid fixture	Route has scrutiny, not EVC	HIGH
T02	Valid scrutiny+EVC category	Reviewer signs required prior state	One eligible EVC task generated	HIGH

ID	Scenario / initial condition	Stimulus	Pass criterion	Priority
T03	Illegal skipped stage	Caller requests EVC completion before scrutiny	Reject 409; unchanged case state	HIGH
T04	Wrong role approval	Ordinary admin attempts statutory decision	Reject 403; audit security event	HIGH
T05	Deficiency return	Initial scrutiny D1 and D2	Applicant correction task with signed snapshot	HIGH
T06	Scope-guard positive	Re-scrutiny edits D1/D2 only	Allowed; previous unflagged items unchanged	HIGH
T07	Scope-guard negative	Re-scrutiny attempts editing cleared C3	Reject with §1.8(g) explanation	HIGH
T08	Repeat correction cycle	Additional permitted re-scrutiny round	Prior rounds preserved; no deletion	MED
T09	Incorrect EVC eligibility	Scrutiny-only application advances	No EVC record created	HIGH
T10	Approval-linked condition	Competent authority enters conditional result	Typed obligation created exactly once	HIGH
T11	NBA conditional 2026 route	Two eligible intake cases with different accreditation	Different verified evaluation paths	HIGH
T12	Boundary X=360	Eligibility fixture hits unexpressed table equality	Policy review required; no speculative route	HIGH
T13	Unit handling of n divisions	Input has no verified division size	No student-seat conversion inferred	HIGH
T14	Session separation	June, January and ODL fixtures	Different source calendars; no universal SLA	HIGH
T15	Appeal origin	EC decision uploaded; repeated request	Timer starts at publication; repeat denied where applicable	HIGH
T16	Officer concurrency	Two updates from same expected_version	Exactly one succeeds; consistent audit/outbox	HIGH
T17	Reminder cancellation	Evidence accepted after alert queued	Stale job canceled or skipped by state recheck	HIGH
T18	Payment evidence distinction	Applicant uploads proof but portal unresolved	Only submitted/pending status shown	MED
T19	Penalty/withdrawal boundary	Calendar cutoff and intimation period interact	Refer to authorized review; no auto-penalty	HIGH
T20	Conditional SHC	Case with no SHC requirement	No hearing auto-created	HIGH
T21	Historical conformance	Seed impossible event sequence	Human-review anomaly with event and rule IDs	MED
T22	Duplicate alert retry	Outbox delivery retried three times	One logical delivered message per dedupe key	HIGH
T23	Expired document after renewal	Renewal recorded with valid period and reviewer pending	Original alert updated; no false verified state	MED
T24	Backdated proof (dual time)	Effective date before upload date	Both actual receipt and claimed validity preserved	LATER
T25	Linked closure→new case	Dependent opening before closure recommendation	Guard prevents premature opening progress	LATER
T26	Rule-version correction	Old case under base APH and new under updated errata	Old case unchanged absent approved migration	HIGH
T27	Evidence integrity	Delete/download link attempted by unauthorized user	403; file confidential; logged	HIGH
T28	Case dashboard accuracy	Transition 3 cases between stages	Counts and ages recalculate from DB	HIGH

19.1 Evaluation metrics

Metric	Definition	Prototype measurement
Routing correctness	Correctly chosen required stages among fixed source-verified test fixtures	Pass/total labeled fixture count
Guard rejection rate	Invalid server transitions correctly blocked	All T03/T04/T07/T09 cases rejected
Audit coverage	Successful state changes with actor, time and source event	Percentage of change events with mandatory fields
Alert reliability	Logical notification once per obligation-version key	Retry/cancel test event counts
Deadline accuracy	Correct due classification using configured source/start	Compare computed due with fixture expected due
Reviewer accountability	Every consequential action has a named authorized actor	Missing-authority count should be zero
Dashboard consistency	Aggregates reflect live event-state changes	Compare SQL counts to dashboard numbers
Administrative task effort	Manual actions in scripted demo before/after, if benchmark defined	Illustrative measure only; not AICTE productivity claim

20. Five-minute demonstrable story and pitch

Video segment	Demo action	Judge takeaway
0:00–0:35	Acknowledge existing AICTE portal; explain multiple rule-dependent case routes	We understand the real problem, not generic forms
0:35–1:15	Submit two fictional cases and open “Why this route?”	Source-linked conditional routing in action
1:15–2:10	Scrutiny flags D1/D2; upload correction; forbidden C3 API mutation fails	Enforceable regulatory re-scrutiny guard
2:10–2:50	Authorized reviewer completes; applicable case creates EVC queue entry	Correct next-stage orchestration
2:50–3:35	Human-authorized demo decision generates compliance obligation	Approval-to-obligation continuity
3:35–4:15	Advance simulated clock; alert fires; submit verification; stale alert disappears	Explainable, exception-safe alerts
4:15–4:45	Show officer dashboard + full event/audit trace + optional conformance flag	Transparency and operational monitoring
4:45–4:55	Summarize impact, limitations and future official integration	Credible closing within five minutes

20.1 Product hero messaging

Headline Every AICTE case, obligation and deadline—with the reason behind the next action.

Subheadline A proposed policy-aware administrative coordination layer that connects approval workflows, re-scrutiny controls, committee tasks and post-approval compliance without substituting for authorized decisions.

Three demonstration proofs Different routing from current source rule; blocked out-of-scope re-scrutiny; approved case becomes tracked compliance with cancelable alerts.

20.2 Questions evaluators might ask

Question	Source-grounded answer
Does AICTE already have a portal?	Yes. Official materials describe online applications, selected automation and communications. We improve source-linked coordination, not claim a first digital portal.
Is the 2026–27 route the same for everyone?	No. Active source clauses, addenda and errata must be evaluated per category/session/accreditation.
What is actually innovative?	Explainable rule provenance plus server-enforced re-scrutiny scope and joined obligation history, demonstrated with functional events.

Question	Source-grounded answer
Can your tool approve an institution?	No; competent-authority decisions remain human-controlled. The prototype records a simulated authorized decision.
Where is your real AICTE API?	Not publicly verified; live access would need formal authorization. The demo uses an explicit mock adapter and fictional cases.
Why no predictive AI?	Publicly verified historical stage event data and benchmark are unavailable; deterministic correctness has higher validity.
What happens if rules change?	Each case carries a source/revision; updates require approved activation and controlled migration rather than silent retrospective edits.

PART VII — Risks, assumptions and final recommendation

21. Risk register

Risk	Severity	Mitigation / fallback
Incorrectly using older 2025–26 rule	High	2026–27 version registry; source and amendment reconciliation test
EVC universally required for intake increase	High	Accreditation-sensitive verified decision matrix; boundary-policy review
Internal officer permissions not public	High	Synthetic role map, deny-by-default and no official-authority claim
Missed or duplicated alerts	High	Outbox with dedupe, current-status recheck and cancel/supersede
Using academic cutoff as officer SLA	High	Separate calendar, category-window and internal policy timers
Document treated as verified on upload	High	Submitted, observed, independently verified distinct states
UI-only lock bypassed through API	High	Server guard using signed snapshot and permitted deficiency IDs
Two officers race to change case state	High	Expected_version optimistic concurrency and atomic audit writes
Assuming public API exists	Medium	No real integration claim; provider-neutral mock/import interface
Claims of novel government feature without internal benchmark	High	Call them proposed differentiators with explicit uncertainty
Excessive formal methods complexity	Medium	Small deterministic checker; defer network solver/predictive AI
Hackathon time pressure	High	Single core journey; phased freeze and local backup demo
Exposure of institutional sensitive files	High	Access scoping, private storage, test-only fictional records
Ambiguous regulations	High	Policy Review Required instead of guessed mandatory decision

22. Final system selection and development handoff

Final recommendation Build an integrated regulation-aware administrative case lifecycle with versioned route explanations, protected re-scrutiny scope, conditional committee tasks, officer-controlled final actions and source-aware obligations. Preserve all four selected research documents as evidence, but this document is the governing team engineering specification for the prototype.

MVP scope freeze Two to three trustworthy route fixtures; signed scrutiny/re-scrutiny; EVC queue; one conditional approval-to-obligation handoff; one expiry/deadline alert with cancellation; stage-age dashboard; complete audit events.

Primary USPs Source-based route transparency and enforceable differential re-scrutiny. Supplement with obligation continuity and exception-aware notification behavior. Optional small conformance module only after the core system is stable.

Not initially Full BPM engine, public-production API adapter, actual AICTE officer access, regulatory legal certificate, predictive ML, sophisticated dynamic temporal constraint solver, broad platform migrations, untested committee-delegation automation.

Next developer action Agree on two source-verified route fixtures and one fictional policy timer; freeze schema and API contracts; create one end-to-end integration test before UI polish; assign the six workstreams from Section 18.

PART VIII — Sources, reconciliation and evidence audit

23. Complete source register

External official and research links below are included as direct references. Where a historic manual or repository is listed through the earlier reports without a fresh independent check of its current service status, this limitation is stated. The bibliography distinguishes regulatory sources from methods and product references. This research does not establish access to authenticated AICTE internal tools.

23.1 Government and regulatory primary sources

G01 — AICTE Approval Process Handbook 2024–2027 — https://aicte.gov.in/sites/default/files/APH%20Final.pdf — APH §§1.8–1.11, 2.23–2.24, Tables 2.2/2.3, Annexures 1 and 7; independently inspected relevant pages.

G02 — AICTE Updated Addendum/Corrigendum 2026–27 (42 pp) — https://aicte.gov.in/downloads/aph/Final%20Addendum%202026-27_Updated.pdf — Updated committee definitions PDF pp.2–3; conditional SHC and §2.25(f) PDF p.11; final errata PDF p.42; independently inspected page images.

G03 — AICTE Academic Calendar 2026–27, dated 2 June 2026 — https://aicte.gov.in/sites/default/files/Academic%20Calendar%202026-27.pdf — Tables A–D; June/January sessions, PGDM/PGCM and ODL distinctions; independently inspected.

G04 — AICTE Addendum/Corrigendum 2025–26 — https://www.aicte.gov.in/sites/default/files/approval/2025-26/APH%20Corrigendum-Addendum.pdf — Historical amendment source; revalidate scope before applying alongside 2026 rules.

G05 — AICTE Approval FAQ 2024–25 — https://aicte.gov.in/sites/default/files/FAQ%20Updated.pdf — Historic existing portal/NSWS context; not current internal UI proof.

G06 — AICTE Security Deposit Payment Manual 2024–25 — https://www.aicte.gov.in/sites/default/files/approval/2024-25/Security_deposit_user_Manual.pdf — Historic portal transaction states and payments; does not confirm bank settlement or production API.

G07 — AICTE 2026–27 Revised Public Notice index (4 Feb 2026) — https://www.aicte.gov.in/node/3342 — Prior report indexed linked notice but PDF returned access restriction; body not revalidated.

G08 — Government PIB launch of 2024–2027 APH, 6 Dec 2023 — https://www.pib.gov.in/PressReleasePage.aspx?PRID=1983013 — Confirms three-year APH publication intent; not a substitute for amended clauses.

G09 — National Single Window System Investor Guide — https://www.nsws.gov.in/s3fs/2021-11/NSWS_Investor_Guide_v3.3.pdf — General government handoff context; not evidence of AICTE approval case-write API.

G10 — AICTE 2026–27 21-page earlier Addendum — https://aicte.gov.in/downloads/aph/Final%20Addendum%202026-27.pdf — Historical earlier addendum; updated 42-page file and errata require precedence review.

23.2 Selected academic research

P01 — Declarative workflows: Balancing between flexibility and support (van der Aalst, Pesic & Schonenberg (2009)) — https://doi.org/10.1007/s00450-009-0057-9 — Constraint-based branches and exceptions; use for case guard design; not drop-in code.

P02 — Correctness criteria for dynamic changes in workflow systems—A survey (Rinderle, Reichert & Dadam (2004)) — https://doi.org/10.1016/j.datak.2004.01.002 — Keep active cases pinned to approved policy versions; controlled migration.

P03 — Event log imperfection patterns for process mining (Suriadi, Andrews, ter Hofstede & Wynn (2017)) — https://doi.org/10.1016/j.is.2016.07.011 — Design event validation and duplicate/out-of-order detection.

P04 — Process Mining Manifesto (van der Aalst et al. (2012)) — https://doi.org/10.1007/978-3-642-28108-2_19 — Define adequate event schema; postpone empirical bottleneck ML pending real data.

P05 — Time patterns for process-aware information systems (Lanz, Weber & Reichert (2014)) — https://doi.org/10.1007/s00766-012-0162-3 — Different clock types, fixed dates, time lags and validity windows.

P06 — Visualization of Compliance Violation in Business Process Models (Awad & Weske (2010)) — https://doi.org/10.1007/978-3-642-12186-9_17 — Explain violated workflow rules and offending event sequences.

P07 — Conformance checking: a state-of-the-art literature review (Dunzer, Stierle, Matzner & Baier (2019)) — https://doi.org/10.1145/3329007.3329014 — A limited deterministic historical-log checker is appropriate.

P08 — YAWL: yet another workflow language (van der Aalst & ter Hofstede (2005)) — https://doi.org/10.1016/j.is.2004.02.002 — Loop/choice/cancellation patterns as validation checklist; full language optional.

P09 — Change patterns and change support features in process-aware information systems (Weber, Reichert & Rinderle-Ma (2008)) — https://doi.org/10.1016/j.datak.2008.05.001 — Support controlled rule and process change rather than silent in-place edits.

23.3 Technical standards and open-source tools

T01 — OMG BPMN 2.0.2 — https://www.omg.org/spec/BPMN/2.0.2/ — General workflow notation; not an AICTE-mandated implementation.

T02 — Flowable engine — https://github.com/flowable/flowable-engine — BPMN/CMMN/DMN candidate; compare operational overhead.

T03 — Temporal — https://github.com/temporalio/temporal — Durable workflow orchestration candidate; not native BPMN tasklist.

T04 — Open Policy Agent — https://github.com/open-policy-agent/opa — General policy engine for future complex permission rules.

T05 — bpmn-js — https://github.com/bpmn-io/bpmn-js — BPMN visualization/editor; review logo/watermark license conditions.

T06 — PM4Py — https://github.com/process-intelligence-solutions/pm4py — Process mining tool; copyleft/license review if embedding; data needed.

T07 — Camunda 7 — https://github.com/camunda/camunda-bpm-platform — Historic comparison; review release/end-of-life status before new adoption.

T08 — PostgreSQL — https://www.postgresql.org/docs/ — Chosen relational database documentation.

T09 — FastAPI — https://fastapi.tiangolo.com/ — Backend option; select against team skills.

T10 — React — https://react.dev/ — Frontend component architecture option.

23.4 Evidence and excluded-material notes

Source G07 remains access-limited, therefore dates or policy changes contained exclusively in its body are unresolved. Source G05/G06 cover older applicant-facing flows and do not certify current internal portal capability. Real AICTE case event datasets and official sandbox/API credentials have not been verified. Legal admissibility is not proven by hash-chain implementation; do not treat general technical literature as regulatory authorization. [B §8; C §I3; D §6]

24. Four-document reconciliation and final audit ledger

The four supplied attachments to this synthesis are: (A) Government-First Research Dossier (14-page source); (B) Targeted Gap-Closure Report (10-page PDF); (C) Final Research Improvement Addendum (4-page source); and (D) Gemini Research: Useful Additions & Corrections (6-page comparison). D references an earlier Gemini dossier of 101 rendered pages, which is not itself one of the four attached input files in this final run. Thus D's extraction was evaluated, but this consolidation is not a claim that the original 101-page Gemini dossier was separately re-audited here.

Issue / prior material	Resolution and final adoption	Why / evidence
A originally selected blanket Scrutiny + EVC for intake increase	CORRECT with 2026–27 accreditation-conditional decision matrix	C and B override older selected APH row; G02 PDF p.42

Issue / prior material	Resolution and final adoption	Why / evidence
A APH-only baseline did not include later amendments	UPDATE versioned source registry with 2025/2026 addenda and errata	B and C cite new effective-scope clauses; G02
Older RO/Chairman committee constitution formulations	CORRECT constituting authority under modified definitions; operational delegation unknown	G02 PDF pp.2–3; B §6
Original unqualified SHC stages	REFINE as “wherever applicable”	G02 §2.23.3; B §3
A generic internal SLA interpretation	REPLACE with fixed/session/event-relative taxonomy, show unknown clock rules	C §C3; D §4; G03
A route and deficiency loop but no server field-scope guard	ADD API-enforced re-scrutiny with signed snapshot and deficiency IDs	D §2; G01 §1.8(g)
A process conformance future idea only	ADD small event-log violation examples; NOT full temporal solver	D §3 and P06/P07
A document history fields already present	REFINE with valid_from/to and received/verified times	D §5; phase full bitemporal later
A estimated risk/due alerts	REFINE with idempotent outbox and stale alert suppression	C §I4; D §4
Gemini advanced CSTNUD dynamic slack model	DEFER; data and rules not available	D §6; absent representative official durations
Gemini cryptographic legal-proof claim	REJECT unconditional certification	D §6; no evidence hash alone establishes admissibility
Gemini broad 30-year and 5-year universal lease claims	QUALIFY by institution, document, amendment and case type	D §6; G01 Annexure 1
Public portal notifications, upload, hybrid EVC, payment	BASELINE, not USPs	G01, G02; B §§4–5
Missing linked dependent cases	OPTIONAL case dependency for simultaneous closure/opening	D §8; G01 §1.8(i)
Current internal UI and production APIs	UNKNOWN / authorization dependent	B §8; C §6
Novelty claims about a government-first product	QUALIFY as proposed domain-focused integration and controlled enhancements	A/B/D warnings; no authenticated competitive audit

24.1 Four documents: unique retained contributions

Input	Distinctive contribution preserved	What was not simply copied
A — baseline research	Problem decomposition, main AS-IS routes, relevant BPM methods, product foundation and MVP architecture	Superseded unqualified intake-EVC rule
B — gap closure	2026 committee changes, current portal evidence, official-access limitations, compliance obligations	Unsupported conclusions about missing private features
C — refinement	X=360 policy review, three timer classes, appeal trigger, payment/evidence states, T11–T20 tests	Unvalidated automatic legal consequences
D — Gemini comparison	Differential re-scrutiny scope guard, explainable conformance, time patterns, dual-time provenance, linked-case edge	Older Gemini 2025 rule baseline, formal solver and legal-certification claims

24.2 Requirements-to-implementation map

PS domain	Mandatory design component	MVP screen/proof	USP relation
A	Policy version + route/guard + deficiency snapshot	Different route + D1/D2/C3 test	Source-linked reasons; scope guard
B	Stage event timestamps + role queries	Pendency and task dashboard	Explainable exceptions
C	EVC eligibility queue + signed report	One scheduled relevant EVC	Correct conditional route
D	Obligation record + evidence review	Approval creates obligation	Lifecycle continuity
E	Document versions + validity + renewal	One expiring document and replacement	Dual-time provenance later

PS domain	Mandatory design component	MVP screen/proof	USP relation
F	Timer jobs + outbox + audit	Alert fires once, cancels after evidence	Exception-aware escalation

24.3 Remaining verification and development checklist

Unknown / unresolved issue	Can prototype continue?	Needed for official deployment
Private AICTE officer interface and current bottleneck dashboards	Yes, with qualified USP wording	Authorized walkthrough and functionality audit
Actual officer role grants and escalation recipients	Yes, with synthetic RBAC and clear labels	Signed role/delegation matrix
Current official integration API + credentials	Yes, using labeled mock adapter	Documented contract and authorized onboarding
Full 2026–27 notice review, including inaccessible revised notice	Yes for carefully selected verified fixtures	Obtain text or official clarification
Intake X=360 and n-division calculation nuance	Yes, enter Policy Review Required	Authoritative interpretation before automating
Official document retention and closure rules	Yes, avoid destructive default deletion	Approved retention and audit policy
Historical EVC/case event logs	Yes, deterministic dashboards	Representative logs if predictive analytics ever proposed

Developer appendix — minimum working contracts

The following examples are internal software-design fixtures, not official regulatory machine-readable rules. Each source-grounded field requires reviewed mapping from the applicable handbook, addendum and case category before production use.

A. Minimal policy object (illustrative)

{ "policy_id": "DEMO-2026-INTAKE-V1", "session": "2026-27", "application_category": "increase_intake", "source": { "document": "G02", "location": "PDF p.42", "review_state": "interpretation_pending" }, "conditions": ["accreditation_valid", "eligible_programme", "threshold_rule_confirmed"], "routes": { "qualifying_accredited": ["scrutiny", "decision_gate"], "non_accredited": ["scrutiny", "evc", "decision_gate"] }, "on_unknown": "policy_review_required" }

This fixture illustrates how a policy engine might distinguish paths; it is NOT a comprehensive codification of the errata, including applicability, source cross-references, special conditions and exception rules. Never activate an illustrative path as official without detailed validation.

B. Server-side guard pseudocode

- load case using submitted expected_version; reject stale case version with 409
- retrieve pinned policy, latest completed stages, actor role and decision authority
- evaluate required prerequisites, applicability and evidentiary status
- for ordinary re-scrutiny: ensure every modified field belongs to signed deficiency_set
- if any guard fails: return explicit rule code without mutating the domain record
- otherwise commit case state + audit event + outbox record atomically
- enqueue EVC or obligation tasks only when accepted event and checked policy justify them

C. Minimum relational invariants

Constraint	Concrete enforcement goal
case row_version	Increment on transition; reject stale expected_version
policy_version_id	Foreign key pins the reviewed revision for case-specific behavior
deficiency snapshot identifier	Re-scrutiny edits checked against saved permitted deficiency IDs
audit_event unique event_id	Prevent duplicate successful state transitions on client retries
timer unique dedupe_key + generation	One logical reminder for obligation/policy generation
outbox event_id unique	One logical side effect; recheck active current state before delivery

Constraint	Concrete enforcement goal
document_version status + time fields	Submission is not independent verification; preserve valid and recorded dates

D. Minimum developer definition of done

A seeded end-to-end test must create a case, resolve a route with a visible source card, sign a scrutiny snapshot, reject an out-of-scope re-scrutiny update, generate only an applicable EVC task, record a role-limited human decision, generate and resolve an obligation, and show a consistent audit and dashboard. Every rejected action should have a machine-readable error code and readable explanation. The test suite must run reliably from a clean seeded database.

Final quality and truthfulness declaration

This document is a consolidated design specification using all four attached reports, with selected independently rechecked primary AICTE pages and academic sources. The implementation is a proposal; no working prototype, production API, actual audit-log tamper-evidence test, security penetration test or AICTE regulatory certification was performed as part of creating this report. The project is ready to move into selected prototype implementation and acceptance testing, subject to explicit unknowns above.

Team sign-off should verify: the two chosen route source clauses, selected demo authority roles, the clearly simulated internal alert calendar, exact UI/API contracts, the ten highest-priority test cases, and the five-minute demonstration story before development begins.